Masterless Muon: Training 4-bit Weights Without a Full-Precision Copy

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Abstract

Low-bit training normally keeps a full-precision copy of every weight. ECO removes the copy by feeding each step's rounding error back into the optimizer's momentum. We show that this fails for Muon: Muon orthogonalises its momentum, so its update has a nearly fixed size, and an error fed into the momentum loses its size and part of its direction on the way to the weights. We propose the *ticker*: each weight is stored only as a 4-bit value, and an 8-bit carry, measured in units of that weight's own grid gap, holds the part of each step that has not yet moved it (about 2.8 bytes per weight with 8-bit momentum). The weights being trained are the 4-bit weights that are served. On GPT-2 small (continued training) the ticker ends within 0.0013 of training with a float copy, and from scratch (45M parameters) within 0.013, while bf16 training rounded to nearest afterwards is 0.04 and 1.16 worse. ECO in its own setting, with AdamW and fp32 moments (8.75 bytes per weight), ends 0.08 behind the ticker on GPT-2 small in each of 3 seeds. In full-parameter RL of Qwen3.5-0.8B on GSM8K with every decoder weight in 4 bits, ours ends 3.7 points behind full-precision training and fails our pre-registered line of -4; most of the gap is the 6.5 points lost by rounding the model to 4 bits before training. Every pass line was registered before its runs, and we report the ones that failed.

1 Introduction

Training a model whose weights are 4-bit numbers almost always means training a different, full-precision model. The standard recipe keeps an fp32 (or bf16) *master copy* of each weight; the optimizer adds its small update to the copy, and the 4-bit weights used in the forward pass are re-rounded from it at every step. The copy is needed because a single step is usually far smaller than the distance between neighbouring 4-bit values: rounded on its own, the step would vanish. The cost is that the model being trained is not the model being served. In reinforcement learning (RL) this shows up directly: the 4-bit generator that writes answers and the full-precision trainer that scores them are different weights, and the mismatch has to be corrected for [16, 26].

Error feedback and why it does not carry over to Muon. ECO [17] removes the master copy for SGD with momentum and for Adam: after each step, the rounding error is added into the momentum, scaled so that the optimizer re-applies it over the following steps. This works because both optimizers are *linear* in the momentum (Adam is linear once its second moment is fixed), so an error put into the momentum comes out in the weights unchanged. Muon [11, 15] is not linear in its momentum. It replaces the momentum by an approximately orthogonal matrix with the same singular vectors (Newton–Schulz iteration), so its update has a nearly fixed size whatever

the momentum's size. We show, with a measurement on pretrained GPT-2 that needs no training (Section 3), that an error fed into Muon's momentum loses its size and part of its direction on the way to the weights.

The ticker. Instead of routing the error through the optimizer, we keep it beside each weight. Each weight stores a 4-bit value and a small integer *carry*: the part of the weight's intended position that the 4-bit grid cannot show yet, in units of the gap to the next grid value on that side. Each step adds the update to weight plus carry, rounds to the nearest grid value, and keeps the rest as the new carry, rounded at random to b bits. With an exact (fp32) carry this is Kahan's compensated summation [12] on 4-bit weights, and it is mathematically the same as training with a float copy. The question is how few bits the carry can have.

Contributions.

1. **The mechanism.** Muon's orthogonalised update has a nearly fixed size, so (i) an error fed into its momentum does not reach the weights intact, which is why ECO fails with Muon, and (ii) the per-step travel of each weight is predictable, which makes a noise model for low-bit carries simple (Section 3).

2. **The ticker**: 4-bit weights, an 8-bit carry in units of the weight's own gap, and 8-bit momentum, about 2.6–2.8 bytes per weight (Section 2). It is within 0.0013 (continued training of GPT-2 small) and 0.013 (45M from scratch) of training with a float copy, within 0.0012 of an exact fp32 carry, and beats bf16 weights with random rounding once both are rounded to nearest on the 4-bit grid, by 0.04–0.05 and by 1.13. ECO run as its paper does, with AdamW, random rounding and fp32 moments (8.75 bytes per weight), ends 0.080 behind it on GPT-2 small in each of 3 seeds.

3. **A noise model** for the smallest carry that works at a given learning rate (Section 4). It *fails* its pre-registered tests on a grid of 4 carry widths and 4 learning rates; the damage depends on the carry width, not the learning rate, and 8 bits was always enough.

4. **Full-parameter RL on 4-bit weights**, with one copy of the weights, on Qwen3.5-0.8B and GSM8K. Ours ends 3.7 points behind full-precision training and fails its pre-registered line (-4); most of the gap is the cost of rounding the starting model to 4 bits. A 16-bit LoRA add-on on the same 4-bit model ends 2.6 points above ours but changes no 4-bit weight: what it learns is stored only in the add-on.

5. **Memory, measured**, against the strongest baseline we know (bf16 weights with random rounding), which ours beats by only 5–13% (0.87–0.95 of its peak); against the standard setup it is 0.32. The pitch of this paper is native 4-bit weights, not memory.

2 Method

2.1 The 4-bit format

Each weight matrix $W \in \mathbb{R}^{m \times n}$ (output $\times$ input) is split into blocks of 16 consecutive weights along the input dimension. A block has one scale $s = \max_i |w_i|/6$, and each weight is $w = s\,x$ with x from the signed E2M1 value set

$$x \in \{0, \pm 0.5, \pm 1, \pm 1.5, \pm 2, \pm 3, \pm 4, \pm 6\},$$

as in NVFP4 [18] (MXFP4 [21] uses the same values with a power-of-two scale per 32 weights). RTN$(\cdot)$ rounds to the nearest grid value. For a grid value x we write $g_+(x)$ and $g_-(x)$ for the gaps to the next value up and down (0.5, 1 or 2 times s; beyond ± 6 we use 2). In our experiments the scale is kept in fp32; NVFP4 stores it in FP8 (E4M3) with one fp32 scale per tensor, which saves 0.19 bytes per weight. Only the matrices inside the transformer blocks are stored this way and trained by Muon; embeddings, norms and biases stay in floating point and are trained by AdamW, as is usual for Muon.

2.2 The ticker

Let $\hat{w}$ be a weight's stored 4-bit value, c its carry, η the learning rate and Δ the optimizer's update for this step. One step is

$$\tilde{w} = \hat{w} + c - \eta\Delta \qquad \text{(where the weight should be)} \tag{1}$$

$$\hat{w}' = \mathrm{RTN}(\tilde{w}) \qquad \text{(nearest 4-bit value; block scale recomputed)} \tag{2}$$

$$r = \tilde{w} - \hat{w}' \qquad \text{(what the grid cannot show yet)} \tag{3}$$

$$k' = \mathrm{SR}\left(2L\,\frac{r}{g_\pm(\hat{w}')}\right) \in \{-L, \ldots, L\} \qquad \text{(the new carry, as an integer)} \tag{4}$$

$$c' = \frac{k'}{2L}\,g_\pm(\hat{w}') \qquad \text{(the value it stands for)} \tag{5}$$

where $g_\pm$ is the gap on the side of r's sign, $L = 2^{b-1} - 1$ for a b-bit carry, and SR rounds to one of the two neighbouring integers at random, with probabilities that make it unbiased. Because $\hat{w}'$ is the *nearest* grid value, $|r| \le g_\pm/2$, so $|k'| \le L$ and the carry never clips. A weight "ticks" to the next grid value when its carry passes half a gap. The carry is relative to the weight's own gap, so a small weight (fine gaps) and a large weight (coarse gaps) both get $2L$ levels of resolution. At $b = 8$ the resolution is $1/254$ of a gap.

Relation to compensated summation. With c kept exactly in fp32, $\hat{w}+c$ equals the float-copy weight at every step, given the same updates: the method is Kahan summation [12] with a 4-bit main term. The ticker quantizes the compensation term, in units that follow the main term's local resolution. Its only error relative to a float copy is the random rounding of the carry (and of the momentum); Section 4 sizes it. Error Diffusion [1] uses the same idea for binary weights: a few-bit residual buffer per weight, updated with stochastic rounding, that flips the weight when it crosses a threshold. The ticker is the multi-level form of that buffer (Section 6).

2.3 8-bit momentum

Muon's momentum is stored as 8-bit integers $(-127 \ldots 127)$ with one fp32 scale per 128 weights of a row, rounded at random after each update. Newton–Schulz runs on the dequantized Nesterov momentum.

2.4 Bytes per weight

Table 1 counts the training state of the block matrices (weights plus optimizer state; gradients and activations are the same for all methods and not counted).

Table 1: Training state per weight of the Muon-trained matrices. Scales: one fp32 scale per 16 weights (0.25 bytes per weight) or, in brackets, NVFP4-style FP8 scales (0.0625). Momentum scale: one fp32 scale per 128 (0.03).

method	parts	bytes/weight
4-bit + float copy, AdamW	fp32 copy + two fp32 moments	12
ECO-AdamW	4-bit + scale + two fp32 moments	8.75
4-bit + float copy, Muon	fp32 copy + fp32 momentum	8
Kahan (exact carry)	4-bit + scale + fp32 carry + 8-bit mom.	5.78
bf16-SR	bf16 weight + 8-bit momentum	3.03
ours, 8-bit carry	4-bit + scale + 8-bit carry + 8-bit mom.	**2.78** (2.59)
naive / ECO-Muon	4-bit + scale + 8-bit mom.	1.78 (1.59)

3 The mechanism: Muon's update has a fixed size

3.1 Muon

For a matrix W with gradient G_t, Muon [11] keeps a momentum $B_t = \beta B_{t-1} + (1-\beta)G_t$ and orthogonalises the Nesterov combination $U_t = (1-\beta)G_t + \beta B_t$:

$$W \leftarrow W - \eta\,\alpha\ \mathrm{NS}(U_t), \qquad \alpha = \sqrt{\max(1, m/n)},$$

where NS is five steps of a quintic Newton–Schulz iteration, $X \leftarrow aX + (bA + cA^2)X$ with $A = XX^\top$, $(a, b, c) = (3.4445, -4.7750, 2.0315)$, started from $X = U/\|U\|_F$. If $U = P\Sigma Q^\top$ is a singular value decomposition, $\mathrm{NS}(U) = Pf(\Sigma/\|U\|_F)Q^\top$ for a fixed odd polynomial f that maps every singular value that is not tiny into roughly $[0.7, 1.2]$. Two facts follow.

Fact 1: the step size is nearly fixed. $\mathrm{NS}(cU) = \mathrm{NS}(U)$ for every $c > 0$, and $\|\mathrm{NS}(U)\|_F \approx \bar{f}\sqrt{\min(m, n)}$ with $\bar{f}$ the root-mean-square of f over the singular values. So each step moves the matrix by $\eta\alpha\bar{f}\sqrt{\min(m,n)}$ in Frobenius norm, or $\eta\alpha\bar{f}/\sqrt{\max(m,n)}$ per weight (root-mean-square), whatever the size of the gradient. On GPT-2 we measure $\bar{f} = 0.905$ (standard deviation 0.066, range 0.69–0.99) over 48 matrices and 40 steps, while the momentum's own RMS varies by a factor of 49 (Figure 1c); multiplying the momentum by 10^3 or 10^{-3} changes the update by less than 0.02%.

Fact 2: an error added to the momentum does not pass through. ECO [17] adds $\alpha_{\mathrm{ECO}}\, e$ to the momentum, with e the rounding error of the last step and $\alpha_{\mathrm{ECO}} = \eta^{-1}(1-\beta^{-1})$ for SGD with momentum (and the same times Adam's per-weight step size for Adam). The coefficient is negative, since $\beta < 1$; the next step subtracts η times the momentum, so the error comes back with the right sign. For an optimizer whose update is a linear function of the momentum, this puts exactly $(1-\beta)e$ into the next step and e into the sum of all later steps, so the weight ends up where the float copy would have put it. Muon's update is not linear in the momentum. To first order, for a small error δ added to U (take $m \le n$), the change in the exact orthogonal factor $PQ^\top$ is

$$d(PQ^\top) = P\,K\,Q^\top + P\,\Sigma^{-1}P^\top\delta\,(I - QQ^\top), \qquad K_{ij} = \frac{\tilde{\delta}_{ij} - \tilde{\delta}_{ji}}{\sigma_i + \sigma_j}, \quad \tilde{\delta} = P^\top\delta Q. \tag{6}$$

Compared with passing δ through at the update's average gain $\kappa = \|\mathrm{NS}(U)\|_F/\|U\|_F$, three things happen:

1. **The symmetric part of $\tilde{\delta}$ is discarded.** It only changes the singular values, and orthogonalisation sets them all to about 1. On GPT-2 this is 22% of the energy of the actual 4-bit rounding error (7–50% by matrix).

2. **The rest is reweighted by $\sigma_{\mathbf{rms}}/\sigma_i$.** Directions with small singular values are amplified and the few large ones, where the gradient's signal lives, are damped. Momentum spectra are heavy-tailed, so the delivered correction points somewhere else than the error. Newton–Schulz caps the amplification: its slope at zero is $a^5 \approx 485$ on the normalised input, which is at most about $485/(\bar{f}\sqrt{\min(m,n)}) \approx 19$ times κ for GPT-2's 768-wide matrices.

3. **Large errors saturate.** By scale invariance $\mathrm{NS}(U+\delta) \to \mathrm{NS}(\delta)$ as $\|\delta\|/\|U\| \to \infty$, so $\|\mathrm{NS}(U+\delta) - \mathrm{NS}(U)\|_F \le 2\bar{f}\sqrt{\min(m,n)}$: the delivered change is bounded, and the share of the intended correction that arrives falls like $\|U\|/\|\delta\|$.

No choice of the injection scale fixes this: a small injection is bent and amplified, a large one is bent and truncated, and in between the gradient's own direction is overwritten.

3.2 Measurement on GPT-2 (no training)

We take pretrained GPT-2 small [20], compute real gradients of its 48 block matrices on FineWeb windows, and build Muon's Nesterov momentum over 40 steps (2 windows of 1,024 tokens each). We then add into U an error whose direction is each matrix's actual 4-bit rounding leftover $W - \mathrm{RTN}(W)$, at sizes from 0.01 to 10 times $\|U\|$, and compare the change in the update with the error (Figure 1, Table 2). For SGD with momentum the error passes through exactly. Adam with ECO's per-weight rule also passes it through exactly, because ECO multiplies the error by the same per-weight factor that Adam divides by; injected without that factor, Adam rescales it weight by weight (cosine 0.80). For Muon, the cosine between the delivered and the intended correction is 0.65–0.77 at every size (single matrices 0.48–0.91), and the share of the intended correction that arrives along the error falls from 7.6 at size 0.01 (small errors are amplified) through 0.90 at size 1 to 0.093 at size 10. At the size ECO's rule gives for this error at learning rate 10^{-3} (22 times the momentum on average), 4.6% of the intended correction arrives, and the update keeps a cosine of only 0.015 with the error-free update: the gradient's direction is gone. Even an error one tenth the size of the momentum turns the update by about 40° (cosine 0.76 with the error-free update). A random error direction behaves the same way, so this is a property of Newton–Schulz, not of the 4-bit leftover. An earlier training-time measurement on a small character-level model found the same cosine, about 0.69, and showed ECO-Muon training ending worse than no error feedback at all (Appendix A).

4 A noise model for the carry

The ticker differs from a float copy only through random rounding of the carry (and of the momentum, which we leave out here). Each step rounds the carry to a multiple of $u_b = g/(2L) \approx g/2^b$, where g is the weight's gap. Stochastic rounding is unbiased, and its error on one step has variance $p(1-p)u_b^2 \le u_b^2/4$. Fresh random bits are drawn at every step, so the errors are independent and after T steps they add up to a random drift of standard deviation at most

$$D_b(T) = \frac{u_b}{2}\sqrt{T} \approx \frac{g}{2^b} \cdot \frac{\sqrt{T}}{2}.$$

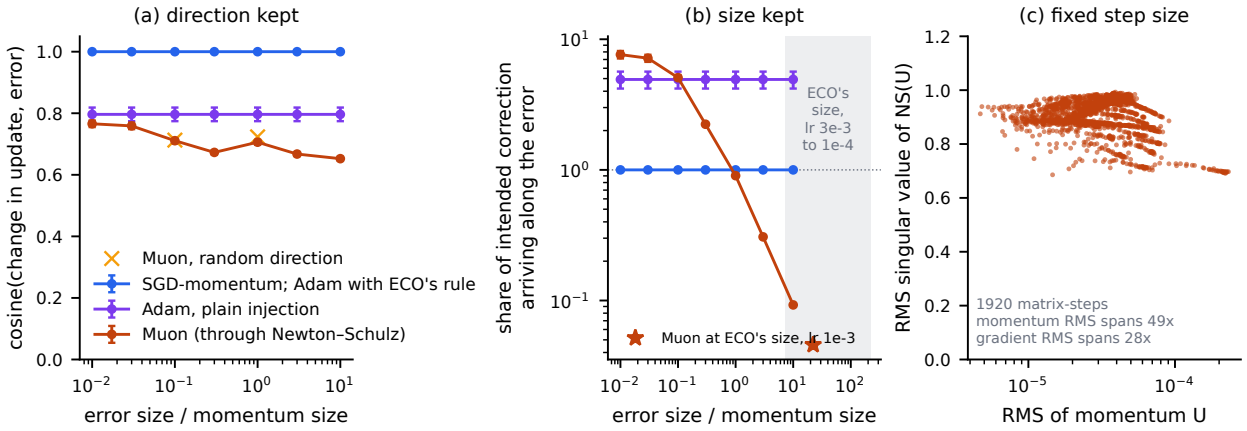


Figure 1: Error fed into the momentum, pretrained GPT-2 small, 48 block matrices (mean ± standard error over matrices). (a) Cosine between the change in the update and the injected error. (b) Share of the intended correction that arrives along the error (= cosine × size ratio; 1 means it passes through intact); the grey band is the size ECO's rule gives at learning rates $3 \cdot 10^{-3}$ to 10^{-4}. (c) Muon's step size: the RMS singular value of NS(U) against the RMS of the momentum U, for every matrix and step.

Meanwhile Muon moves each weight by about $\eta\,\alpha\bar{f}/\sqrt{\max(m,n)} =: \eta\,\bar{u}$ per step (Fact 1). If the weight keeps moving in one direction its intended travel is $\eta\bar{u}T$, and the ratio

$$\rho = \frac{D_b(T)}{\eta\,\bar{u}\,T} = \frac{g}{2^{b+1}\,\eta\,\bar{u}\,\sqrt{T}}$$

measures how much of that travel is noise. A carry of b bits is enough when ρ is below a threshold ρ^*, which we fit on two learning rates and test on two others (Section 5.3).

Assumptions, stated honestly. (i) Rounding errors at different steps are independent; this holds for the random bits, but the weight's position also feeds back into the gradient. (ii) Travel is linear in T. If the update's sign flips from step to step, the travel grows more slowly (like $\sqrt{T}$ for a pure random walk), and ρ underestimates the damage. (iii) The gap g is fixed; in reality it changes when a weight moves to another grid cell or the block's largest weight changes the scale. (iv) The momentum's 8-bit rounding noise is ignored. (v) The model predicts a per-weight noise level, not the loss: mapping ρ to a loss gap assumes the damage grows with ρ (P3a tests only the ranking). The model is cleaner for Muon than for Adam because Fact 1 makes $\bar{u}$ a constant of the matrix shape; under Adam the per-weight step $|\hat{m}|/\sqrt{\hat{v}}$ varies with each weight's gradient signal-to-noise ratio.

5 Experiments

All pass lines were written before the runs (`PAPER_PLAN.md`, revision 3) and are reported whether or not they hold. Within a table all arms run on the same GPU type and number format. Seeds: 3 for close arms, 1 for arms expected to be far apart; a 1-seed gap counts only if it exceeds 10 times the largest seed standard deviation of the same table. Every loss carries a paired standard error over 256 fixed validation windows of 1,024 tokens.

Table 2: Survival of an error added to the momentum (actual 4-bit rounding leftover as direction). Mean over the 48 block matrices of GPT-2 small, momentum built over 40 steps (`runs/paper/M_RESULT.md`). Share arriving: the part of the change along the error, as a share of the error passed through at the optimizer's average gain.

optimizer	cosine	size ratio	share arriving along the error
SGD-momentum	1	1	1
Adam, ECO's rule	1	1	1
Adam, plain injection	0.80	7.3	4.9
Muon, error = 0.1× momentum	0.71	7.1	5.1
Muon, error = 1× momentum	0.71	1.28	0.90
Muon, error = 10× momentum	0.65	0.142	0.093
Muon, ECO's size at $\eta = 10^{-3}$ (22×)	0.65	0.071	0.046

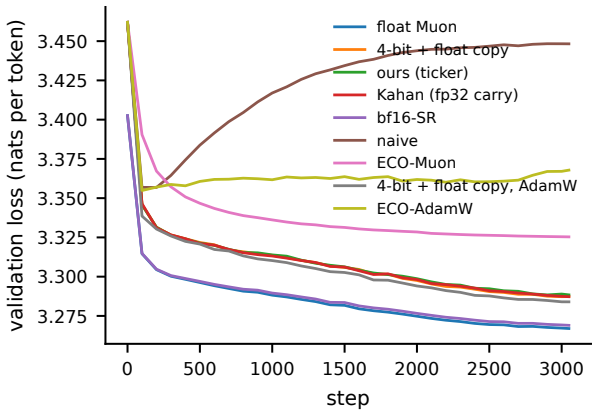


Figure 2: P1: validation loss of every arm, seed 1. Ours, Kahan and the 4-bit float copy overlap; naive rounding gets worse after the first steps; ECO-Muon and ECO-AdamW stall.

5.1 P1: GPT-2 small, continued training (headline)

Pretrained GPT-2 small [20], 100M FineWeb [19] tokens (3,050 steps of 32 × 1024 tokens), Muon at 10^{-3} on the 48 block matrices and AdamW at 10^{-4} on the rest, 5% warm-up then cosine to 10%. One A100, fp16 with loss scaling, TF32 off, deterministic. Arms: ours (8-bit carry, 8-bit momentum), 4-bit with a float copy and Muon, bf16 weights with random rounding and 8-bit momentum (bf16-SR), Kahan (ours with an exact fp32 carry), float Muon, 4-bit with a float copy and AdamW, naive (4-bit weights rounded at random, no carry), ECO-Muon (ECO's injection into Muon's momentum) and ECO-AdamW (ECO as its paper runs it, see below). Table 3 gives the final losses; all five pass lines hold.

What it shows. With 2.78 bytes per weight, ours ends 0.0013 (mean of 3 seeds) behind training the same 4-bit model with an 8-byte float copy, and 0.0012 behind an exact fp32 carry, so the 8-bit carry loses almost nothing. Training in 4 bits beats training in higher precision and rounding to nearest at the end: bf16-SR is the best model while it is scored in bf16, but rounded to nearest 4-bit values it is 0.043–0.049 worse than ours, and float Muon rounded to 4 bits is 0.042 worse. The price of 4-bit weights themselves (ours against float Muon in fp32) is +0.021, the same as for the float copy (+0.020). ECO-Muon does better than naive rounding but freezes the 4-bit weights:

Table 3: P1: final validation loss, GPT-2 small, 100M tokens. ±: paired standard error over 256 windows (evaluation noise). Seed standard deviation (largest 3-seed arm) 0.0033, so a 1-seed gap counts only above 0.033.

arm	bytes/weight	seed 1	seed 2	seed 3
ours	2.78	3.2884	3.2892	3.2889
4-bit + float copy, Muon	8	3.2873	3.2886	3.2866
bf16-SR, rounded to 4 bits	3.03	3.3346	3.3387	3.3322
bf16-SR, scored in bf16	3.03	3.2690	3.2720	3.2695
Kahan (fp32 carry)	5.78	3.2872		
float Muon	8	3.2670		
float Muon, rounded to 4 bits	8	3.3303		
4-bit + float copy, AdamW	12	3.2840	3.2859	3.2846
ECO-AdamW (random rounding)	8.75	3.3679	3.3693	3.3687
naive (random rounding, no carry)	1.78	3.4483		
ECO-Muon (Newton–Schulz in fp32)	1.78	3.3253		
start: pretrained (float)		3.4025		
start: pretrained, rounded to 4 bits		3.4620		

P1a: ours − float copy = $+0.0011, +0.0006, +0.0023$ (± 0.0005); line $+0.03$	PASS
P1b: naive − ours = 0.160, ECO-Muon − ours = 0.037 (± 0.002); line 0.02 and 0.033	PASS
P1c: bf16-SR rounded − ours = $0.046, 0.049, 0.043$ (± 0.001)	PASS
P1d: ours − Kahan = $+0.0012 \pm 0.0005$; line 0.01	PASS
P1e: ECO-AdamW − ours = $0.079, 0.080, 0.080$ (± 0.0014); line 2× s.e.	PASS

only 0.0004% of them change per step (ours: 0.14%), falling to zero by the end. The injected error dominates the momentum, so each step mostly points back towards undoing the last rounding (Section 3). Its first run, with Newton–Schulz in fp16 like every other arm, overflowed by step 10, because the injected error makes the momentum far larger than usual; the run in Table 3 repeats it with Newton–Schulz in fp32, which can only help it.

ECO in its own setting (P1e). ECO-Muon is our adaptation; the ECO paper gives SGD and Adam. P1e runs its Adam version [17, Algorithm 3]: the same 4-bit grid, random rounding (which the ECO paper finds works best), the rounding error injected into Adam's first moment, both moments in fp32, and the learning rate, betas and weight decay of the AdamW float copy, untuned for ECO. This arm and its pass line were added after an outside review and before the runs (Appendix B). ECO-AdamW ends 0.080 behind ours in every seed, and 0.084 behind the float copy trained with the same AdamW; ours is 0.0040 behind that float copy at 2.78 bytes per weight against 12. It drops to 3.359 in the first 300 steps and then stays flat while the float copy keeps improving. Its weights are not frozen: 1.6% of its 4-bit values change per step (ours: 0.14%). A likely reading, not measured: each AdamW step is much smaller than a 4-bit gap, so random rounding adds noise of about a gap at every step, and the injection returns only a $(1 - \beta_1)$ share of each step's error to the weights; the ticker keeps all of it.

5.2 P2: from scratch, 45M parameters

A GPT-2-style model with 6 layers of width 512 (45.2M parameters) is trained from random weights on 0.9B FineWeb tokens (27,466 steps of 32×1024, about 20 tokens per parameter). The Muon learning rate (0.04, with AdamW at $4 \cdot 10^{-3}$) was picked from three by short runs of the float copy, the same rate for every arm. Ours and the 4-bit float copy run 3 seeds, bf16-SR one.

Table 4: P2: final validation loss, 45M model from scratch, 0.9B tokens.

arm	bytes/weight	seed 1	seed 2	seed 3	mean ± seed s.d.
ours	2.78	3.9127	3.9262	3.9196	3.9195 ± 0.0067
4-bit + float copy, Muon	8	3.9078	3.9133	3.9095	3.9102 ± 0.0028
bf16-SR, scored in bf16	3.03	3.8894			
bf16-SR, rounded to 4 bits	3.03	5.0515			
P2a: ours − float copy = +0.0049, +0.0128, +0.0101 (±0.0019); line +0.03					PASS
P2c: bf16-SR rounded − ours = 1.132; line 2 × 0.0067					PASS

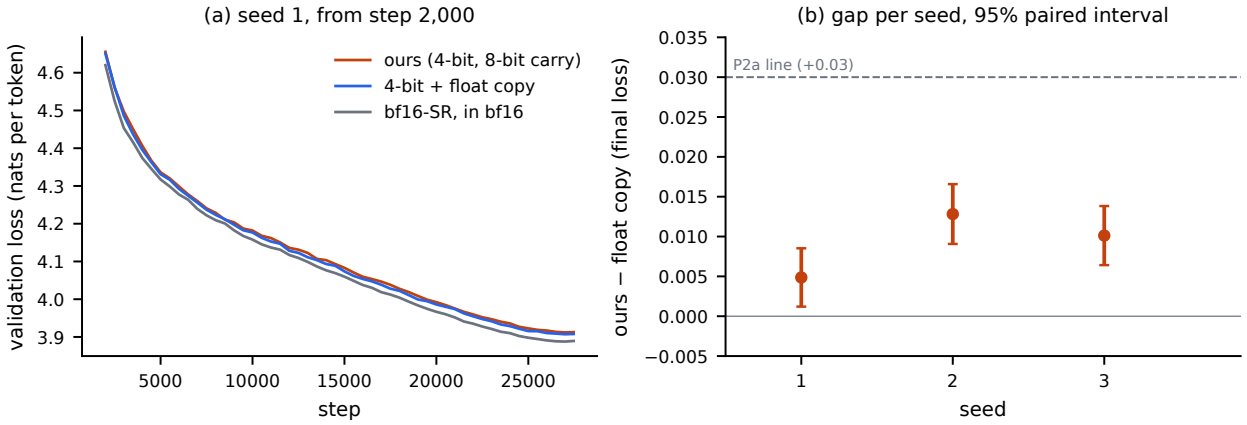


Figure 3: P2: (a) validation loss from step 2,000, seed 1; (b) ours minus the float copy at the end, per seed.

What it shows. From scratch the gap to the float copy is larger than in continued training (mean +0.009 against +0.0013) and resolved on every seed, but well inside the line, and the carry never overflowed. Ours also varies more from seed to seed (0.0067 against 0.0028): the random rounding of the carry adds a little seed noise. Training in 16 bits and rounding to nearest at the end does not survive from scratch: bf16-SR is the best model in bf16 (3.889) and loses 1.16 when rounded to nearest 4-bit values. Post-training methods that round more carefully, such as GPTQ, were not tried and would likely recover part of this loss. The chosen learning rate is the top of its grid, so a larger one might be better for every arm.

5.3 P3: the noise model

Continued training of GPT-2 small for 10M tokens (305 steps, same schedule shape), at Muon learning rates $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}\}$, with carries of $\{4, 6, 8, 12\}$ bits and the float copy at each rate (20 runs, seed 1). Table 5 gives each cell's gap to the float copy and Figure 4 plots it against ρ. **Both pass lines fail.** The rank correlation between ρ and the gap is 0.795 (P3a, line 0.8). The threshold fitted on the two middle rates, $\rho^* = 1.01$, separates working from failing cells there without error, but on the held-out rates it predicts 12 bits at 10^{-4}, where 6 bits already worked (P3b: two levels off).

Why it fails, and what holds instead. The damage depends on the carry width and hardly on the learning rate: 4 bits costs about 0.045 at every rate, 6 bits 0.002–0.004, and 8 and 12 bits nothing measurable. The model assumes each step adds a fixed amount of rounding noise, so small

Table 5: P3: ours minus the float copy at the same learning rate, 10M tokens, seed 1 (±: paired standard error).

carry bits	10^{-4}	$3 \cdot 10^{-4}$	10^{-3}	$3 \cdot 10^{-3}$
4	$+0.0425 \pm 0.0013$	$+0.0518 \pm 0.0012$	$+0.0513 \pm 0.0012$	$+0.0452 \pm 0.0010$
6	-0.0013 ± 0.0004	$+0.0017 \pm 0.0005$	$+0.0041 \pm 0.0005$	$+0.0022 \pm 0.0006$
8	-0.0005 ± 0.0002	-0.0002 ± 0.0003	-0.0000 ± 0.0003	-0.0006 ± 0.0005
12	-0.0001 ± 0.0001	$+0.0001 \pm 0.0002$	$+0.0004 \pm 0.0003$	$+0.0007 \pm 0.0005$

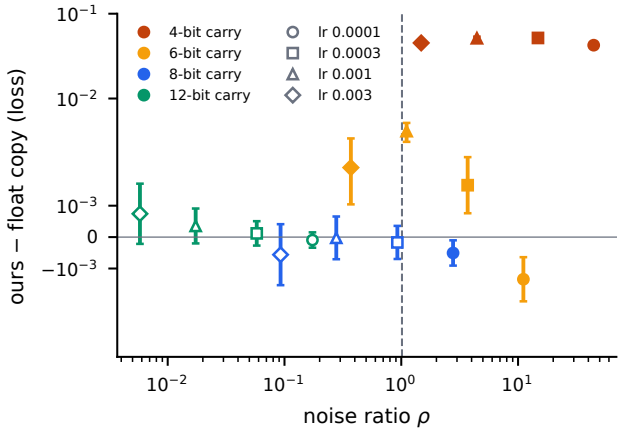


Figure 4: P3: gap to the float copy against the noise ratio ρ (bars: 2 paired standard errors; open markers: within 2 errors of zero). Dashed: the fitted threshold ρ^*.

learning rates, which need many small steps, should suffer most. They do not. Random rounding of a small change adds noise in proportion to that change (its variance is about $u_b\,|\text{step}|$, not $u_b^2/4$), so the noise shrinks with the step. The carry width alone ranks the cells with a correlation of 0.92; this is a reading after the data, not a tested claim. In practice an 8-bit carry was enough at every learning rate we tried, and the RL experiments use it. Before any RL result, the failed model predicted a 12-bit carry for P5; that prediction was recorded and not tested.

5.4 P4: ablations

Starting from ours, one thing is changed at a time (GPT-2 small, 25M tokens, seed 1; looks, not pass lines).

Rounding the weight to nearest matters most: with random rounding a third of the weights flip every step and the loss is 0.071 worse. Measuring the carry in units of each weight's own gap helps a little (0.0012). Error Diffusion's per-step cap never binds at this learning rate, since Muon's steps are far below a quarter of a gap.

5.5 P5: full-parameter RL in 4 bits

Setup. Qwen3.5-0.8B (post-trained, non-thinking mode), GSM8K [2], GRPO [23] with 8 answers to each of 32 questions per step, reward 1 if the final number is right, 200 steps, no KL term, one update per step. Answers are written by vLLM in the same process, with the trained weights copied in (in bf16) after every step. The exam is the 1,319 test questions, greedy, at the start and the end. Three arms: (a) full-parameter training in bf16 with an fp32 copy, Muon on the decoder

Table 6: P4: one change at a time (GPT-2 small, 25M tokens, seed 1).

run	final loss	minus ours	changing per step
ours	3.3047		0.133%
carry in absolute units (block's largest gap)	3.3059	$+0.0012 \pm 0.0005$	0.101%
weight rounded at random, not to nearest	3.3759	$+0.0713 \pm 0.0012$	31.2%
Error Diffusion's cap (0.25 gap per step)	3.3043	-0.0004 ± 0.0004	0.133%

Table 7: P5: GSM8K test accuracy (%), Qwen3.5-0.8B, 200 RL steps. Intervals: 95%, each question's result averaged over seeds and paired across arms.

arm	weights as scored	start	end (per seed)	gain
(a) bf16 + fp32 copy	bf16	54.7	72.9, 71.1, 72.9	$+17.6 \pm 2.4$
(b) 4-bit frozen + LoRA	4-bit + bf16 add-on	48.1	70.6, 70.2, 72.8	$+23.1 \pm 2.5$
(b) merged, rounded to 4 bits	4-bit	48.1	–, 47.6, 47.8	-0.4 ± 2.0
(c) ours	**4-bit**	48.1	69.7, 67.7, 68.4	$+20.4 \pm 2.4$
P5a: (c) − (a) = −3.7, interval $[-5.2, -2.2]$; line: lower end above −4				FAIL
P5b: (c) gain = +20.4, interval $[+18.0, +22.9]$; line: ≥ 5, above 0				PASS
measured: (c) − (b) $= -2.6 \pm 1.6$; (c) − (b) rounded to 4 bits $= +20.9 \pm 2.2$				
measured: 4-bit start − float start $= -6.5 \pm 2.8$				

matrices; (b) the 4-bit model frozen, with LoRA [9] of rank 32 on every decoder matrix, trained with AdamW at 10^{-5} (QeRL-style [10]); (c) ours: every decoder matrix stored in 4 bits with an 8-bit carry, Muon. (b) and (c) start from the same round-to-nearest 4-bit model. Muon's learning rate was 10^{-4} with a guard written in advance; the guard fired ((a)'s training reward peaked near step 35 and collapsed, and its exam score fell from 54.7% to 36.6%), so as written every run uses $4 \cdot 10^{-5}$.

What it shows. Ours ends 3.7 points behind full-precision training and fails P5a. Most of that gap is the 4-bit start: rounding the model costs 6.5 points before any training. Ours gains 20.4 points against 17.6 for (a), but a model recovering from rounding has an easier start than one improving from a higher score, so this does not show that ours learns better. Every weight does learn while stored in 4 bits: 0.11% of the 4-bit values change per step. The generator and the trainer compute with the same numbers: both round each 4-bit value times its fp32 block scale to bf16 in the same way (with FP8 block scales, as in the release, that product is exact in bf16). The remaining gap between their token probabilities (mean 0.0033 for ours, 0.0030 for (a)) comes from the two programs' arithmetic, not from rounding. Against LoRA, ours learns faster early (Appendix C), and LoRA ends 2.6 points ahead after 200 steps (interval $[-4.2, -1.0]$ for ours minus LoRA). The comparison is not like for like: (b) is scored with its add-on in 16 bits, as QLoRA-style methods are served, while ours is scored with 4-bit weights only. To measure what LoRA's gain is worth as a 4-bit model, at the end of (b)'s seeds 2 and 3 we merged the add-on into the 4-bit weights and rounded to the same grid, block scales recomputed. Before these runs we wrote down the prediction that most weights would round back to their start and the model would score near its 48.1% start. It held more strongly than written: the add-on averages 0.5% of the gap between neighbouring 4-bit values, *no* 4-bit value changes, and the rounded model scores 47.7%, 23.8 ± 2.3 points below the same model with its add-on and 20.9 ± 2.2 below ours. This is a statement about where the learning is stored, not a comparison of how LoRA is used: QLoRA-style models are served with

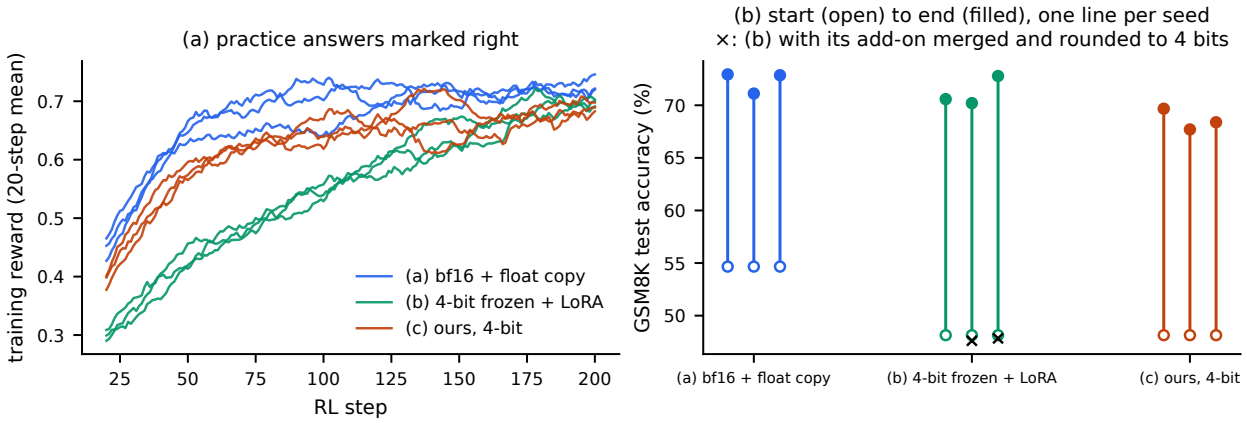


Figure 5: P5: (a) training reward, 20-step running mean, every run; (b) GSM8K test accuracy at the start and after 200 steps, one line per seed; ×: (b) with its add-on merged and rounded to 4 bits (seeds 2–3).

Table 8: P6: memory of one full-parameter training step of Qwen3.5-0.8B on a T4 (GiB).

setup	peak	weights + state	peak / standard
bf16 + AdamW with fp32 copy (standard)	11.74	9.83	1.00
float Muon	10.33	6.56	0.88
bf16-SR, 8-bit momentum	4.31	2.40	0.37
bf16-SR, updated layer by layer	3.96	2.40	0.34
ours	**3.74**	**2.18**	**0.318**

the add-on, not merged and rounded. LoRA's gain is real but held in 16 bits; ours is held in the 4-bit weights. Peak GPU memory in these runs is set by the generator's reserved share and the activations (27–30 GB for (a) and (c), 24 GB for (b), which needs no gradients or optimizer state for the frozen weights); the 4-bit values are held in fp32 tensors here, so P6, not P5, measures the method's memory.

5.6 P6: memory, measured

`p6_memory.py` uses the release optimizer with real packed storage (two 4-bit codes per byte, FP8 block scales, int8 carry, int8 momentum), applies the update to each matrix during the backward pass and frees its gradient at once, and checkpoints activations. It measures one full-parameter training step of Qwen3.5-0.8B (752M parameters, of which 498M in the decoder matrices) on a T4 with a batch of 2×1024 tokens.

Ours peaks at 0.318 of the standard setup (P6a, line 0.35: PASS). Against bf16-SR the saving is small: 0.87 of its peak, 0.95 against bf16-SR with the same layer-by-layer update, and 0.91 in steady state. The decoder matrices alone shrink by $2.59/3.03 = 0.86$, but the embedding is a third of this model and is stored the same way in both. The update adds 4.8 s to a 13.5 s step in plain, unfused PyTorch. As a demonstration, a full-parameter step of Qwen3.5-2B fits on the T4 at 8.34 GiB in our format, where the standard setup runs out of memory.

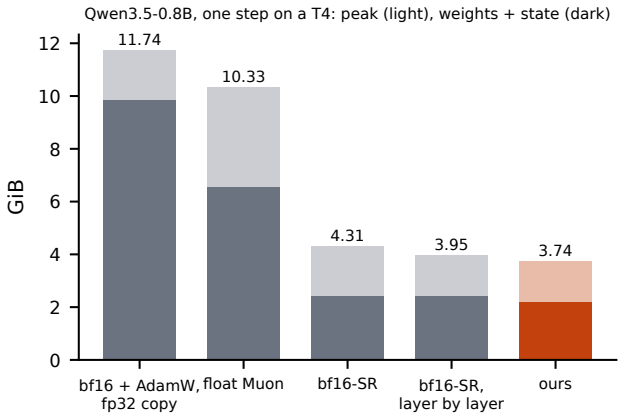


Figure 6: P6: peak memory (light) and weights plus optimizer state (dark) of one training step.

6 Related work

Training without master weights. ECO [17] removes the master copy for SGD with momentum and Adam by injecting the rounding error into the momentum; it proves exact equivalence to master weights for SGD with momentum under an exact (memory-costly) rule and gives a noise-floor bound for the memory-free rule. We show why the same injection fails for Muon (Section 3). *Error Diffusion* [1], which appeared days before this work, replaces the latent weight of a binary network with a 2–8-bit residual buffer updated by stochastic rounding, flips a weight when the buffer crosses a threshold, and reports that the method works best when the buffer grid has a level exactly at that threshold; it trains binary language models of up to about 19M binary parameters. Our ticker is the multi-level form of that buffer: the threshold becomes half the gap to the neighbouring 4-bit value on each side, and the buffer is measured in units of that gap. The parts that are new here are Muon and the explanation of why error feedback through its momentum fails, 4-bit (E2M1) grids with per-block scales, GPT-2-scale experiments, the noise model, and full-parameter RL. Bop [8] argued earlier that binary networks need no latent weights at all and trains them with a momentum-threshold rule.

Compensated summation and stochastic rounding. Kahan summation [12] carries the low-order part lost in each addition. Zamirai et al. [28] show that pure bf16 training matches fp32 if weight updates use stochastic rounding or Kahan summation, and Collage [27] represents weights as several low-precision components to avoid fp32 copies. Stochastic rounding for deep learning goes back at least to Gupta et al. [7]; Croci et al. [3] survey its theory and uses. Our bf16-SR baseline and our naive 4-bit baseline are of this kind. The ticker differs by keeping the compensation term per weight, in few bits, and in units of the main term's own local resolution.

Error feedback in compressed optimization. Carrying a compression error forward to later steps is the core of 1-bit SGD [22] and of error-feedback methods for sign and compressed gradients [13]. There the error lives in the gradient communicated between machines; here it lives in the weights.

Muon and low-bit Muon. Muon [11] orthogonalises the momentum of each matrix with a Newton–Schulz iteration; Liu et al. [15] scale it to large language models. MuonQ [24] and 4-bit

Muon with GRASP [25] shrink Muon's *optimizer state* to 4 bits, both noting that orthogonalisation amplifies quantization error in the momentum (GRASP locates it in the top singular subspace); they keep full-precision weights. Our measurement is the weight-side counterpart: error deliberately fed into the momentum does not come out intact.

4-bit training. NVFP4 [18] and the microscaling formats [21] define the 4-bit values and block scales we use. Full-Stack FP4 [5] pretrains with 4-bit projections, optimizer states and attention but keeps bf16 master weights. FlashOptim [6] reduces optimizer memory and splits the master weight into lower-precision parts with a bound on the resulting error, so a bf16-level master weight still exists.

RL with quantized weights. QLoRA [4] fine-tunes a frozen 4-bit model through a 16-bit low-rank add-on [9]; QeRL [10] applies this to RL with an NVFP4 model, and our arm (b) follows it. The trained model is then the 4-bit base plus a 16-bit add-on: QeRL's own evaluation attaches the adapter to the NVFP4 model in vLLM, or merges it into a 16-bit model. Merging the add-on into the 4-bit weights and rounding again can erase the adaptation, because the nearest grid point is usually the base itself; Scale-QLoRA [14] reports losses of up to 39 points and avoids them by training only the block scales and freezing every 4-bit code. Ours goes the other way: every 4-bit code may change, and the trained model is the 4-bit model, with no add-on to merge. In our LoRA arm the merge-and-round loss is total: no 4-bit value changes and the whole 23.8-point gain is lost (Section 5.5). FlashRL [16] generates rollouts with a quantized copy and corrects the mismatch with the trainer by truncated importance sampling [26]. With the ticker, the trainer and the generator compute with the same bf16 values of the 4-bit weights, so there is no rounding mismatch between them (other arithmetic differences remain).

7 Limitations

- **Scale.** The largest training runs are GPT-2 small (124M, continued training) and a 45M model from scratch; RL uses Qwen3.5-0.8B. Behaviour at billions of parameters is not tested.
- **Memory.** Against bf16 weights with random rounding and 8-bit momentum, ours saves only 5–13% at the peak. Against a frozen 4-bit model with LoRA it uses more training memory, since every weight needs a gradient and a carry. The case for the method is native 4-bit weights trained in full, not memory.
- **RL.** One model (0.8B), one task, 200 steps, three seeds. Ours ends 3.7 points behind bf16 training and fails its pre-registered line; whether the gap keeps closing with longer training, or with a short healing run before RL, is not tested. LoRA's learning rate was not tuned. The generator holds a bf16 copy of the 4-bit values.
- **Simulation.** Training runs keep the 4-bit values in fp32 tensors and do not pack codes; only P6 measures real packed storage. Block scales are fp32 in training; FP8 scales are counted, not trained with (a small character-level test found them to cost little; Appendix A).
- **The mechanism measurement** uses one model, one optimizer state built over a short window of steps, and one error direction (the rounding leftover of the pretrained weights) plus a random control; it does not follow the error through many steps of training.

- **The noise model** failed both of its tests (Section 5.3); the alternative reading (noise in proportion to the step) was made after seeing the data and is not tested.
- **Only weights** are 4-bit. Activations and gradients stay in 16/32 bits; embeddings, norms and biases are not quantized.

8 Conclusion

Error feedback through the momentum, which removes the full-precision copy for SGD and Adam, does not work for Muon, because Muon's orthogonalised update has a nearly fixed size. Keeping the error beside each weight instead, as a carry of 8 bits in units of the weight's own grid gap, trains 4-bit weights with no copy at about 2.8 bytes per weight and ends within 0.0013 (continued training) and 0.013 (from scratch) of training with a copy; ECO with AdamW, at three times the memory, ends 0.08 behind it. In RL the method trains every weight of a 0.8B model in 4 bits, but ends 3.7 points behind full-precision training, mostly because of the cost of rounding the starting model. The open questions are scale, longer RL runs, and whether a short healing run before RL closes the remaining gap.

Code and data. The optimizer (with packed 4-bit storage), every experiment script, the pre-registration with its dated change log, and the log of every run in this paper are at `https://github.com/NibblesOn/masterless`.

Use of AI tools. The experiment code, the analysis scripts and drafts of this text were written with substantial help from an AI coding assistant (Claude, by Anthropic), working under the author's direction. The author set the questions and pass lines, checked the results, and is responsible for all content.

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A Earlier small-scale results

These runs led to the method. They use a character-level GPT on Shakespeare (4 layers, width 128, 0.8M parameters, 2,000 steps; loss in nats per character, mean ± standard error over 3 seeds) and are not evidence for any main claim. Numbers are from `runs/m1`–`m4`, `runs/m7`, `runs/ticker` and `runs/twobit` (`RESULT.md` in each).

Muon breaks ECO (milestones 1–4, CPU). Float Muon reached 1.6278±0.0032 and AdamW 1.9150±0.0034. 4-bit weights with a float copy (Muon) reached 1.6463±0.0019; without a copy, with random rounding, 1.7636 ± 0.0044. ECO with Adam reached 1.9283 ± 0.0039, close to its own float AdamW, as the ECO paper reports. ECO's injection adapted to Muon (error into the momentum before Newton–Schulz) reached 1.8981 ± 0.0057, *worse* than no feedback at all. Measured during training, the change Newton–Schulz made in response to the injected error had a cosine of about 0.69 with the error (all steps, each of 3 seeds: 0.691, 0.692, 0.692), and the error grew from 0.15 to 0.50 of the momentum's size as the learning rate decayed. Adding the error after Newton–Schulz instead, from a 2-byte buffer, reached 1.6733 ± 0.0043.

The ticker (side study T, Colab T4, fp16). With a 4-bit carry (2.28 bytes per weight) the ticker reached 1.6501 ± 0.0045, against 1.6458 ± 0.0035 for 4-bit with a float copy (8 bytes), and beat a 4-bit buffer with random rounding of the weight (1.7120 ± 0.0019) and FP8 training rounded to 4 bits afterwards (1.6717 ± 0.0043) in every seed.

2-bit weights (side study 2B). With 2-bit weights an 8-bit carry reached 1.7476 ± 0.0059 against 1.7253 ± 0.0015 for a float copy with Muon; a 4-bit carry reached 1.8027 ± 0.0040.

A first GPT-2 run with a 4-bit carry (side study G, seed 1, T4, fp16). Continued training of GPT-2 small for 100M tokens with a 4-bit carry ended $+0.0465 \pm 0.0010$ above 4-bit with a float copy, missing a 0.03 line. The cause was a flaw of the 4-bit carry: after a tick the leftover is about half a gap, the carry's largest value, so weights park on the rounding line and flip with the sign of each update; 1.6–1.9% of the 4-bit values changed at every step regardless of the learning rate. In a probe with random gradients, an 8-bit carry removed the effect. This is why the main experiments use an 8-bit carry.

B Changes to the pre-registered plan

The plan, its pass lines and every change are in `PAPER_PLAN.md`, dated, in the repository. No pass line was moved after a result it judges was seen. Table 9 lists every change that touches what was run or how it is judged.

Table 9: Changes to the plan, in order. "Before" means before any run that the change affects.

date	change	when, and why
10-28, rev. 2–3	P5a: "mean of 2 seeds at least (a) −2 points" became "lower end of the 95% interval of (c) − (a) above −4"; P5 re-scoped; P6 cut to 1.5B; bf16-SR baseline added	before any run; outside feedback
10-28, rev. 3	RL model Qwen2.5-0.5B-Instruct became Qwen3.5-0.8B, and P6's model became Qwen3.5-0.8B on a T4; rule for P5's third seeds added; 2-bit arms and 124M from scratch dropped	before any run; the owner's request and compute
10-28	three details made precise by the code (Kahan arm's bytes 5.78, not 4.78; the G0 checks)	before any P0–P6 run
10-28	RL carry: 8 bits, not the 12 bits the noise model predicted (P3c); P3c is therefore not tested	after P3, before P5; the noise model had failed both tests
10-29	questions per RL step 64 became 32, by the plan's own pilot re-costing rule; optimizer Muon (plan's rule); learning rate 10^{-4} with a guard written in advance	after the pilot, before P5
10-29	P5 switched to a 30-step design to save compute, then back to the 200-step design of rev. 3; the 30-step design was completed and is reported (Appendix C)	one 30-step result of (c) was read before the switch back (disclosed)
10-29	the learning-rate guard fired at 10^{-4}, so every 200-step run uses $4 \cdot 10^{-5}$, as written	the guard read training reward only
10-29	third seeds of (a) and (c), by the rule; LoRA seeds 2–3 and LoRA merged and rounded to 4 bits (measured only, prediction written)	after seeds 1–2, before these runs
10-29	ECO with AdamW added to P1 with a new pass line, P1e	after an outside review, before these runs

C The 30-step RL design

For a time the plan judged P5 on a shorter design: 30 steps × 64 questions × 8 answers, Muon 10^{-4}. It was completed after the switch back to 200 steps and is reported here in full; it is not used

for any verdict. Arm (a) seed 1 of this design was the best of three learning rates on the test set, which favours (a).

Table 10: The 30-step design: GSM8K test accuracy (%), Qwen3.5-0.8B. Intervals as in Table 7.

arm	start	end (per seed)
(a) bf16 + fp32 copy	54.7	70.3, 71.9
(b) 4-bit frozen + LoRA	48.1	52.4
(c) ours	48.1	62.5, 66.6
(c) − (a) = −6.60, interval [−8.53, −4.66] (as a P5a line: would fail)		
(c) gain = +16.38, interval [+13.74, +19.01]; (a) gain = +16.45 ± 2.48		
(c) − (b) = +12.13 ± 2.54		